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CLASS:P5

C PROGRAMMINGWEEK-9

CODE

```c
//PES1UG25AM095

#include<stdio.h>
#include<stdlib.h>

int main()
{
int old_n;
int* x;
printf("Enter number of elements: ");
scanf("%d",&old_n);

x=(int *)malloc(old_n*sizeof(int));
if(x==NULL)
{
printf("Memory allocation failed!!!!\n");
free(x);
}

printf("Enter %d elements for malloc:\n",old_n);
for(int i=0;i<old_n;i++)
{
scanf("%d",(x+i));
}
printf("Elements  stored using malloc:\n");
for(int i=0;i<old_n;i++)
{
printf("%d ",*(x+i));
}

int *y;
y=(int *)calloc(old_n,sizeof(int));
printf("\nEnter elements in calloc:\n");
for(int i=0;i<old_n;i++)
{
scanf("%d",(y+i));
}
printf("\nElements stored using calloc:\n");
for(int i=0;i<old_n;i++)
{
printf("%d ",*(y+i));
}

int new_n;
printf("\nEnter new size for realloc: ");
scanf("%d",&new_n);
int *z;

z=(int *)realloc(y,new_n*sizeof(int));
if(z==NULL)
{
printf("Memory allocation failed!!!!\n");
free(z);
}
free(x);
free(y);
y=NULL;
printf("Enter %d elements after realloc:\n",new_n-old_n);
for(int i=old_n;i<new_n;i++)
{
scanf("%d",(z+i));
}
printf("Elements  stored using realloc:\n");
for(int i=0;i<new_n;i++)
{
printf("%d ",*(z+i));
}

free(z);
printf("\nMemory deallocated successfully using free()");
z=NULL;
x=NULL;
}
```

PROGRAM1

OUTPUT

```
Enter number of elements: 10
Enter 10 elements for malloc:
1 2 3 4 5 6 7 8 9 10
Elements  stored using malloc:
1 2 3 4 5 6 7 8 9 10
Enter elements in calloc:
1 2 3 4 5 6 7 8 9 10

Elements stored using calloc:
1 2 3 4 5 6 7 8 9 10
Enter new size for realloc: 15
Enter 5 elements after realloc:
11 12 13 14 15
Elements  stored using realloc:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
PS C:\Users\Chethan S\Documents\C PROGRAM BASICS CON
```

CODE

```c
//PES1UG25AM095


#include<stdio.h>
#include<stdlib.h>

int main()
{
int old_n;
float total=0;
float* expense;
printf("Enter number of months: ");
scanf("%d",&old_n);

expense=(float *)malloc(old_n*sizeof(float));
if(expense==NULL)
{
printf("Memory allocation failed!!!!\n");
free(expense);
return 1;
}

printf("Enter the expense for %d months:\n",old_n);
for(int i=0;i<old_n;i++)
{
scanf("%f",(expense+i));
total+=*(expense+i);
}
printf("Expenses entered are:\n");
for(int i=0;i<old_n;i++)
{
printf("Month %d: %.2f\n",i+1,*(expense+i));
}
printf("Total expense for %d months = %.2f\n",old_n,total);
int new_n,extra_n;
printf("\nEnter new total number of elements: ");
scanf("%d",&new_n);
extra_n=new_n-old_n;
float *new_expense;

new_expense=(float *)realloc(expense,new_n*sizeof(float));
if(new_expense==NULL)
{
printf("Memory allocation failed!!!!\n");
free(new_expense);
return 1;
}
expense=NULL;
printf("Enter the expense for additional %d months\n",extra_n);
for(int i=old_n;i<new_n;i++)
{
scanf("%f",(new_expense+i));
total+=*(new_expense+i);
}
printf("Updated list of expenses:\n");
for(int i=0;i<new_n;i++)
{
printf("Month %d: %f\n",i+1,*(new_expense+i));
}
printf("New total expense for %d months = %.2f\n",new_n,total);
free(new_expense);
new_expense=NULL;
free(expense);
printf("\n\nMemory deallocated successfully using free()");

return 0;
}
```

PROGRAM 2

OUTPUT

```
PS C:\Users\Chethan S\Documents\C PROGRAM BASICS CONCE
WEEK-WISE\WEEK-9\" ; if ($?) { gcc expense_using_dynam
Enter number of months: 6
Enter the expense for 6 months:
1100 1300 1900 2100 1500 1200
Expenses entered are:
Month 1: 1100.00
Month 2: 1300.00
Month 3: 1900.00
Month 4: 2100.00
Month 5: 1500.00
Month 6: 1200.00
Total expense for 6 months = 9100.00

Enter new total number of elements: 10
Enter the expense for additional 4 months
2200 2000 1100 2400
Updated list of expenses:
Month 1: 1100.00
Month 2: 1300.00
Month 3: 1900.00
Month 4: 2100.00
Month 5: 1500.00
Month 6: 1200.00
Month 7: 2200.00
Month 8: 2000.00
Month 9: 1100.00
Month 10: 2400.00
New total expense for 10 months = 16800.00


Memory deallocated successfully using free()
PS C:\Users\Chethan S\Documents\C PROGRAM BASICS CONCE
```